Motivation Statement

Why HackerFab IITB

I am interested in joining HackerFab IITB because it brings together two areas I want to explore deeply: semiconductor fabrication and making advanced engineering technologies more accessible. I find the idea of developing open-source tools for hands-on semiconductor education particularly exciting because it bridges the gap between theoretical concepts and practical engineering. Semiconductor devices are at the heart of modern electronics, yet the processes required to manufacture and characterize them are often difficult for students to access. I would like to be part of an initiative that helps make these processes more approachable through affordable tools, reproducible methods, and shared technical knowledge.

What I Hope to Learn and Contribute

I want to develop a practical understanding of the complete fabrication workflow, including lithography, thin-film deposition, thermal processing, patterning, and electrical characterization. Beyond learning individual processes, I am interested in understanding how they interact and how decisions at one stage affect the final device. I hope to contribute through systematic documentation, careful experimentation, structured process planning, and a willingness to learn from failures and iterate on designs. I also value clear documentation because reproducibility is essential when building tools and processes that others should be able to use.

My Interest in Semiconductor Fabrication

What interests me most is the transition from a designed structure to a measurable electrical device. Patterning a material stack, establishing electrical contacts, and analysing its response requires an engineer to connect material properties, geometry, process limitations, and measurement data. I find this intersection of fabrication and experimentation especially engaging. This is why I selected a simple MOS-capacitor test structure for my accompanying proposal. It offers a focused way to explore fabrication and characterization challenges while keeping the scope realistic, rather than attempting to solve the much broader challenge of fabricating a complete transistor.

Technical Background and Project Experience

My engineering interests include electronics, sensors, measurement systems, and hardware prototyping. Through my work on an Arduino-based automated density-measurement system, I have gained experience connecting electronic components, acquiring sensor readings, calibrating measurements, and integrating hardware with programmed control. This project has strengthened my appreciation for systematic testing, troubleshooting, and validating results against expected values. I hope to apply this experimental mindset while developing the additional skills required for semiconductor tool development and process integration.

What I Dream of Building

In the long term, I would like to help develop an affordable, well-documented, teaching-scale semiconductor-fabrication workflow that enables students to investigate device physics through direct experimentation. My goal is not to replicate an industrial fabrication facility, but to help create a practical learning platform where the relationship between materials, processes, equipment, and electrical behaviour can be explored openly. I see HackerFab IITB as an opportunity to learn collaboratively, contribute to meaningful hardware development, and help make semiconductor fabrication more accessible to future engineers.